

		$2.5 \text{ m s}^{-1} + 2.5 \text{ m s}^{-1} = 5.0 \text{ m s}^{-1}$
5	A	By conservation of momentum $m (4v) + 3m(-v) = (4m) V_{\text{final}}$ $V_{\text{final}} = 0.25 v$
6	C	Tension in the cord = mg